MULTIMODAL AI HACKATHON 2026 · TRACK B: MAPPING FLOOD DAMAGE FROM SPACE

HimaRoad. Flood Copilot

A situation-report copilot fed by our own flood segmentation model. Satellite evidence for flood mapping, cut-off settlement screening and bilingual situation reports in Himalayan valleys, where one road can decide who gets help.

AI component: situation-report copilot (English + नेपाली) · Own UNet flood model, trained from scratch · Free data only: Sentinel-1, Sentinel-2, Copernicus DEM, OSM · Team HimaRoad

"A system that shows what the satellite saw, what it infers, and what it could not see."

Case study: Bhote Koshi and Trishuli valley, Rasuwa, Nepal. Real Copernicus DEM, pre-event OpenStreetMap roads (orange), settlements (white) and the DEM flow path (light blue band).

- **22** / 73 — settlements potentially road-cut in the primary scenario (S2, hypothetical)
- **102** — potentially exposed buildings: overlap with a hazard, not damage
- **121** — suspected affected road links; operational status stays unknown
- **94.0%** — of the area observed for change after fusing 3 radar passes
- **27.0 km** — DEM flow path traced; 21 settlements inside its 500 m corridor
- **86.7%** — UNet development F1 (rule baseline 75.8%)
- **0** / 1,298 — new-water pixels the UNet found on the blind Himalayan test. Reported as is.
- **32** / 32 — predefined copilot cases pass with exact English/Nepali claim parity
- **165** — Nepali wording entries reviewed by a native speaker
- **62,263** — evidence claims verified for the Trishuli run, each with a source and a hash

A The challenge and the system

What problem we solve, how the parts fit together, and which data may enter

01 THE PROBLEM

In a Himalayan valley, one road can decide who gets help

- On **26 Aug 2026** a debris flood moved down the Bhote Koshi and Trishuli valley in Nepal.
- One trunk road and a few bridges connect the villages to the nearest town, health post and hospital.
- Clouds block optical satellites. Radar sees through clouds, but steep slopes hide parts of the ground.
- A relief team needs two answers: **which settlements may be cut off, and what is the evidence?**

~60 h — from the event date to the first Sentinel-1 pass; event time unknown

36 days — until the first clear Sentinel-2 image (1 Oct 2026)

73 — settlements mapped in the area from pre-event OSM

Three research questions

RQ1 Can a model trained only on open flood datasets map new water in the same Sentinel-1 gamma0 domain that it sees at inference?

RQ2 Can surface-change evidence become settlement-level access results that keep **"unknown"** separate from **"safe"**?

RQ3 Can a copilot answer rescuers in English and Nepali **without inventing a number**?

Event facts come from Copernicus and CDSE catalogue metadata. The system never states casualty figures or the cause of the event.

02 WHAT THE SYSTEM DOES

Give an area and a flood date. Get three answers.

1 · Where did the flood hit?
Compares Sentinel-1 radar before and after the event. Maps possible new water and debris-like change. Adds Sentinel-2 optical when the sky is clear.

2 · What was damaged?
Overlays pre-event OSM buildings, roads and bridges. Features that overlap a hazard are **potentially exposed**. Overlap is not confirmed damage.

3 · Who is cut off?
Builds road and path graphs. Finds settlements that can lose their link to the nearest town or health facility, in four **hypothetical** scenarios.

Bonus · Where does the water go next?
From any upstream point, traces a flow path down the valley on the Copernicus DEM and lists the settlements along it.

AI component: the situation-report copilot. It answers rescuers' questions and writes one-page situation reports in English and Nepali. **Every number comes from the system's maps.** Our own UNet (trained from scratch on Kuro Siwo) and the radar change detector give it its water evidence.

Four contributions

C1 · Compliant pipeline
Raw Sentinel-1/2, GLO-30 and pre-event OSM to a run bundle with full provenance.

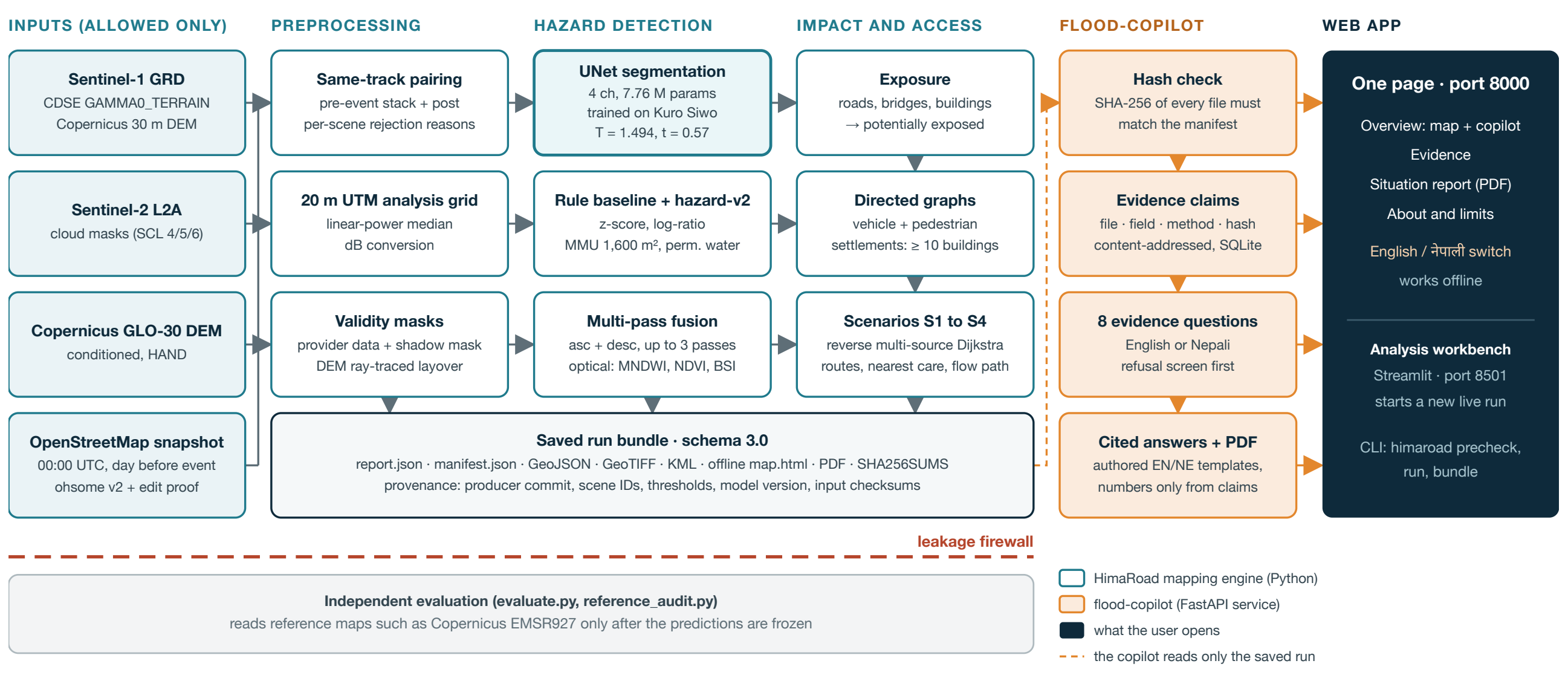
C2 · Gamma0-domain UNet
Trained from scratch, pre-registered, calibrated to a 0.80 precision floor.

C3 · Evidence-aware access
Four hypothetical scenarios. Unobservable means unknown, never zero.

C4 · Grounded bilingual copilot
Predefined registered questions. Every number comes from the maps.

03 SYSTEM OVERVIEW

One system: from raw satellite data to a cited situation report



INPUTS (ALLOWED ONLY)
- **Sentinel-1 GRD** — CDSE GAMMA0_TERRAIN, Copernicus 30 m DEM
- **Sentinel-2 L2A** — cloud masks (SCL 4/5/6)
- **Copernicus GLO-30 DEM** — conditioned, HAND
- **OpenStreetMap snapshot** — 00:00 UTC, day before event, ohsome v2 + edit proof

PREPROCESSING
- **Same-track pairing** — pre-event stack + post, per-scene rejection reasons
- **20 m UTM analysis grid** — linear-power median, dB conversion
- **Validity masks** — provider data + shadow mask, DEM ray-traced layover

HAZARD DETECTION
- **UNet segmentation** — 4 ch, 7.76 M params, trained on Kuro Siwo, T = 1.494, t = 0.57
- **Rule baseline + hazard-v2** — z-score, log-ratio, MMU 1,600 m², perm. water
- **Multi-pass fusion** — asc + desc, up to 3 passes, optical: MNDWI, NDVI, BSI

IMPACT AND ACCESS
- **Exposure** — roads, bridges, buildings → potentially exposed
- **Directed graphs** — vehicle + pedestrian, settlements: ≥ 10 buildings
- **Scenarios S1 to S4** — reverse multi-source Dijkstra, routes, nearest care, flow path

Saved run bundle · schema 3.0
report.json · manifest.json · GeoJSON · GeoTIFF · KML · offline map.html · PDF · SHA256SUMS
provenance: producer commit, scene IDs, thresholds, model version, input checksums

FLOOD-COPILOT
- **Hash check** — SHA-256 of every file must match the manifest
- **Evidence claims** — file · field · method · hash, content-addressed, SQLite
- **8 evidence questions** — English or Nepali, refusal screen first
- **Cited answers + PDF** — authored EN/NE templates, numbers only from claims

WEB APP
- **One page · port 8000** — Overview: map + copilot, Evidence, Situation report (PDF), About and limits, English / नेपाली switch, works offline
- **Analysis workbench** — Streamlit · port 8501, starts a new live run
- CLI: himaroad precheck, run, bundle

leakage firewall

Independent evaluation (evaluate.py, reference_audit.py)
reads reference maps such as Copernicus EMSR927 only after the predictions are frozen

- ☐ HimaRoad mapping engine (Python)
- 🟧 flood-copilot (FastAPI service)
- ■ what the user opens
- - - the copilot reads only the saved run

04 DATA INTEGRITY

A strict data firewall keeps every result honest

MAY ENTER INFERENCE
- Sentinel-1 (CDSE gamma0)
- Sentinel-2 L2A
- Copernicus DEM GLO-30
- OSM from before the event
- Kuro Siwo (training)
- Sen1Floods11: surveyed, not used

VALIDATION ONLY, AFTER FREEZE
- Copernicus EMS (EMSR927)
- UNOSAT products
- GFM flood maps
- Post-event OSM edits

The freeze, in time order

1. **02:17:40 IST:** case-study runs frozen, tag `case-study-frozen`.
2. **02:17:51.87 IST:** the reference folder appears, after the freeze.
3. Every read of reference material is logged in `LEAKAGE_LOG.md`.
4. Nothing was tuned on the reference after the comparison.

Enforced in code
`cdse.fetch` refuses every collection except Sentinel-1 GRD and Sentinel-2 L2A. Only `evaluate.py` and `reference_audit.py` read reference material.

Enforced in time
OSM snapshot at 00:00 UTC on the day before the event (2026-08-25T00:00Z), with an edit-timestamp proof.

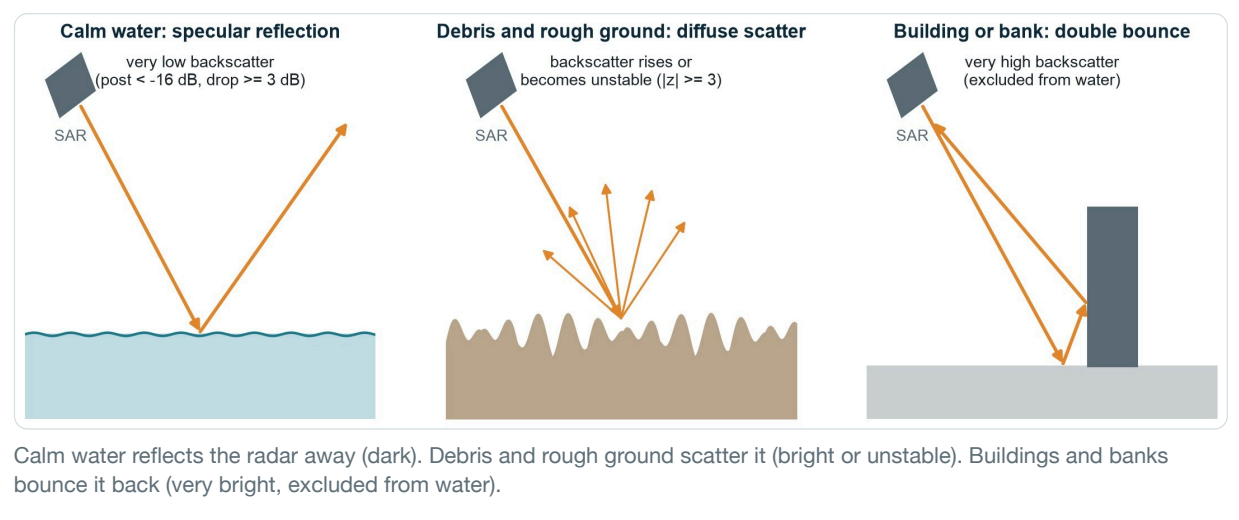
No threshold, model, hyperparameter or checkpoint is chosen from reference content. Earth Engine is a flagged fallback; it was not used for the case study or for judging.

B Where did the flood hit?

Change detection from Sentinel-1 radar, with honest observability

05 RADAR PHYSICS

Change detection that follows radar physics


Calm water: specular reflection
very low backscatter (post < -16 dB, drop >= 3 dB)
SAR
Debris and rough ground: diffuse scatter
backscatter rises or becomes unstable (|z| >= 3)
SAR
Building or bank: double bounce
very high backscatter (excluded from water)
SAR

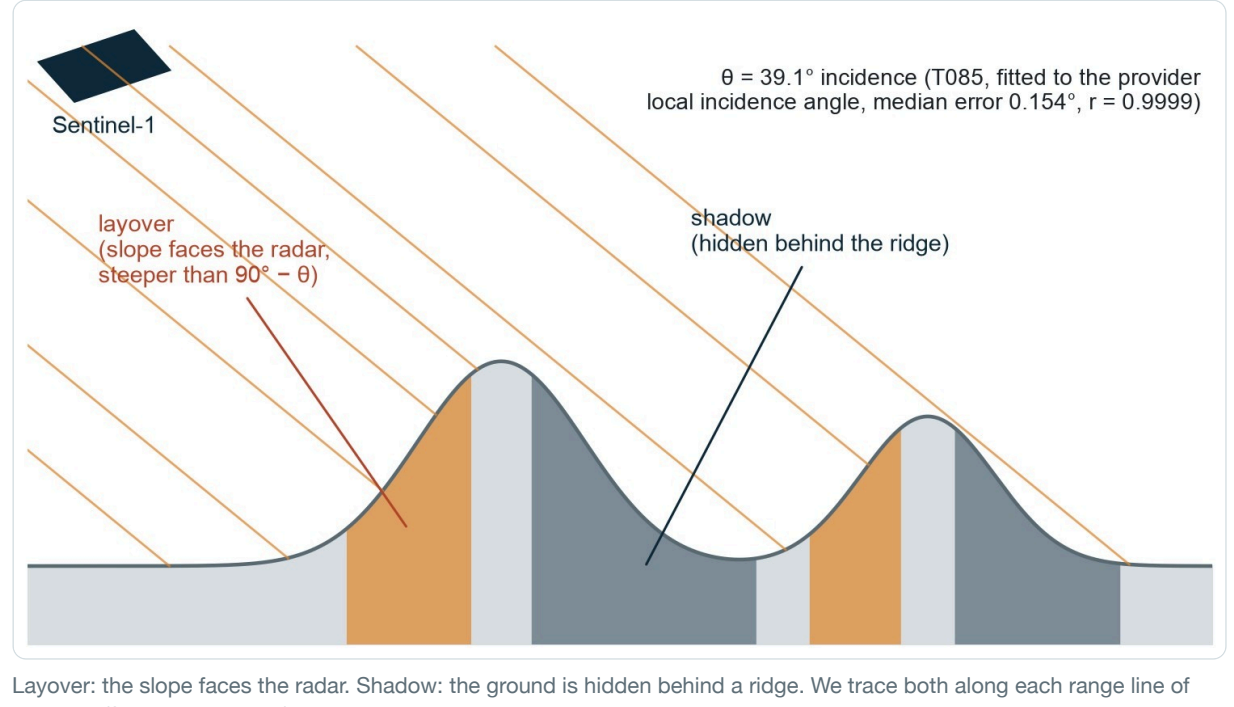
Calm water reflects the radar away (dark). Debris and rough ground scatter it (bright or unstable). Buildings and banks bounce it back (very bright, excluded from water).

- **New water:** post-event $\sigma^0 < -16$ dB *and* a drop $\Delta \ge 3$ dB from the pre-event median of the same-track stack (linear power).
- **Disturbance:** $z = (\sigma^0_{post} - \mu_{pre}) / \max(\sigma_{pre}, 1\ \text{dB})$, $|z| \ge 3$, with at least 3 pre-event dates. A two-sided test finds darkening and brightening.
- **Possible debris:** brightening in the valley-bottom corridor (HAND ≤ 30 m). **Unvalidated** rule evidence: it can also be a landslide, scour, construction or vegetation change.
- **Filters:** permanent water out (pre VV < −20 dB, persistence 0.5); slope > 12° suppressed; polygons under 1,600 m² (4 px) removed.
- **Terrain and optics:** HAND ≤ 20 m is a soft flag only (20/30/50 m sensitivity recorded). Sentinel-2 indices give separate corroboration.

All thresholds are in `configs/default.yaml` and are written into every output.

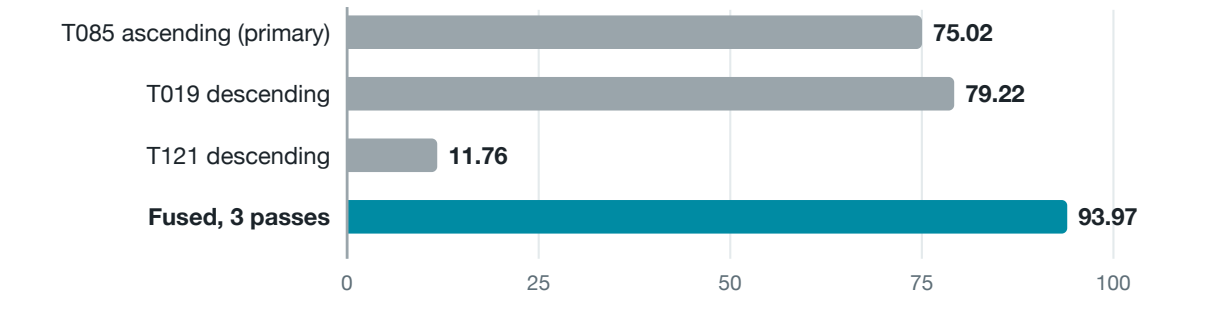
06 STEEP TERRAIN

Layover and shadow hide valley walls, so we fuse three passes


Sentinel-1
θ = 39.1° incidence (T085, fitted to the provider local incidence angle, median error 0.154°, r = 0.9999)
layover (slope faces the radar, steeper than 90° − θ)
shadow (hidden behind the ridge)

Layover: the slope faces the radar. Shadow: the ground is hidden behind a ridge. We trace both along each range line of the DEM (θ = 39.1° on T085) and add the provider's shadow mask.

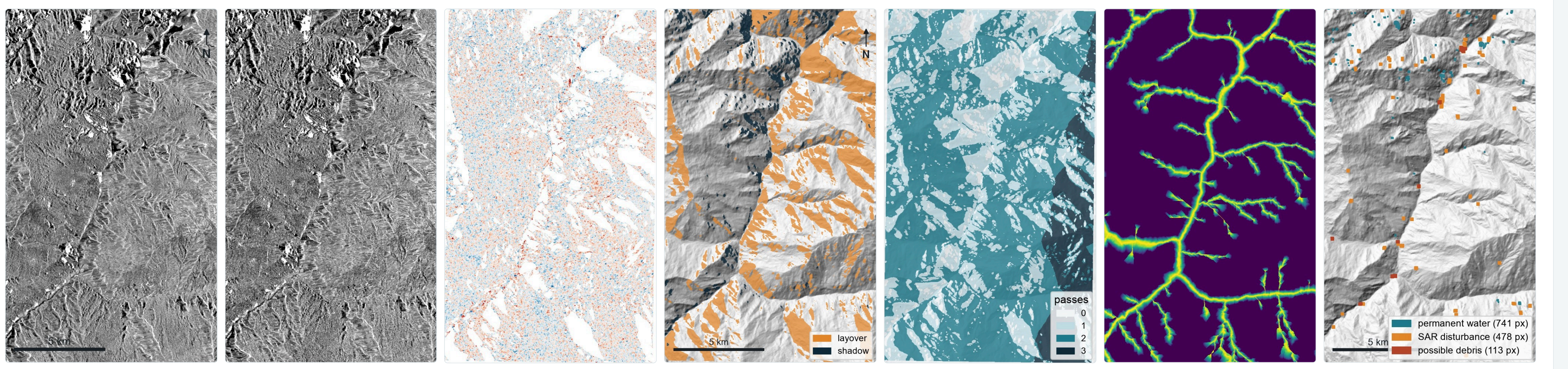
Valid change observation per pass, Trishuli (%)



Unobservable means unknown, never zero. 24.9% of the area is masked by layover or shadow, and 56.1% of road links are unobservable. They stay "unknown", not "safe".

07 REAL DATA · TRISHULI VALLEY · FROZEN RUN R2

What the radar saw, what the terrain hides, and the hazard layers we built



1 · Before Pre-event VV median of 3 dates (gamma0, dB)
2 · After Post-event VV, 28 Aug 2026 12:21 UTC
3 · Change z-score: blue darker, red brighter, white masked
4 · Hidden ground T085 layover (orange, 21.8%) and shadow (navy)
5 · Fusion passes that observe each pixel (0 to 3)
6 · Terrain HAND from GLO-30: yellow = valley floor (soft flag)
7 · Hazard layers permanent water, SAR disturbance, possible debris

Sensing the valley
3 orbit tracks (T085 asc, T121 desc, T019 desc) · 11 + 3 pre/post Sentinel-1 scenes · 12 candidate scenes rejected, each with a reason.

How the images align
Gamma0 processed on request from raw GRD by CDSE Sentinel Hub (`GAMMA0_TERRAIN`, `COPERNICUS_30`). One north-up 20 m UTM grid. Per-date provider masks define valid data.

Pairing rule
Only scenes from the same orbit track are compared. With no same-track pair, the run ends as `incomplete` with the reason. It never draws a map that only looks valid.

Satellite time is not event time
Area 85.30–85.42° E, 28.12–28.30° N. Radar before: 23 Jul, 4 Aug, 16 Aug 2026 (same track). After: 28 Aug 2026 12:21 UTC, orbit 85 ascending,
60.4 h
after the event-date boundary. Valid post-event observation 99.08%.

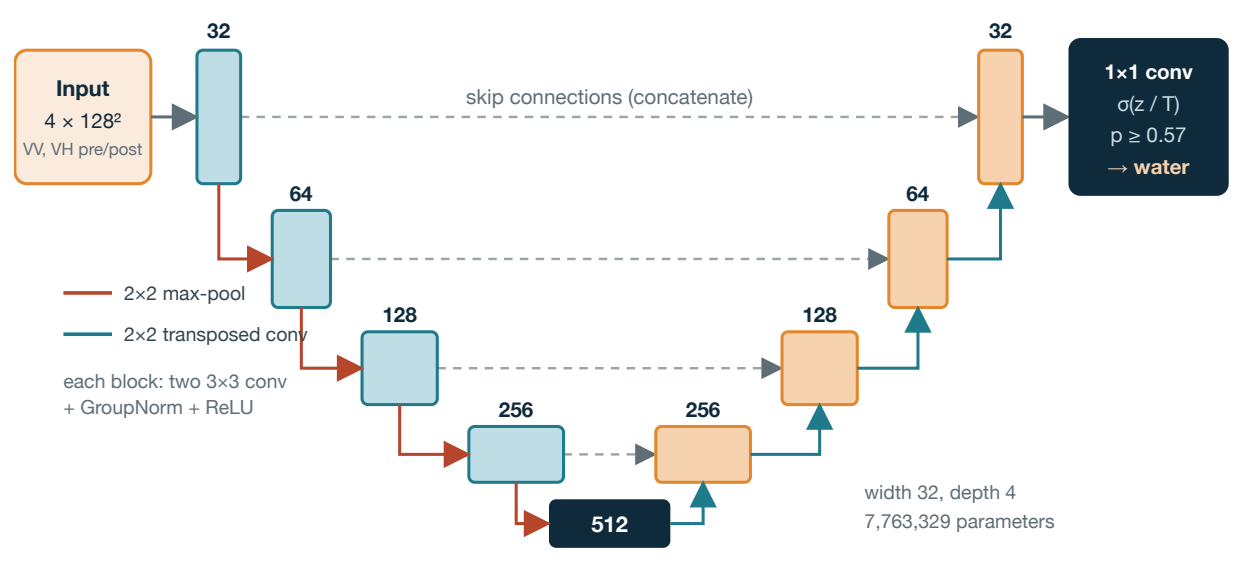
Source: report.json of the frozen Trishuli run r2 (producer 23a2527, schema 3.0, hazard-v2, tag `case-study-frozen`). Contains modified Copernicus Sentinel data 2026.

C Our AI, part 1: a UNet flood segmentation model

Trained from scratch, pre-registered, calibrated, and tested once on held-out Himalayan floods

08 THE SEGMENTATION MODEL

A UNet trained from scratch in the inference domain



7.76 M — parameters, no pretrained encoder

4 ch — pre-event VV/VH median and post-event VV/VH

20 m — UTM grid, the same CDSE gamma0 as inference

1.7 s — GPU inference per 600 km² (4.9 s on CPU)

- Tiled inference: 128 px tiles, 16 px overlap, logits averaged.
- Water head only. **There is no debris prediction head.**
- The model gives the copilot its water evidence, together with the rule change detector.

09 TRAINING DATA AND PROTOCOL

Event-disjoint data and a registered training pipeline

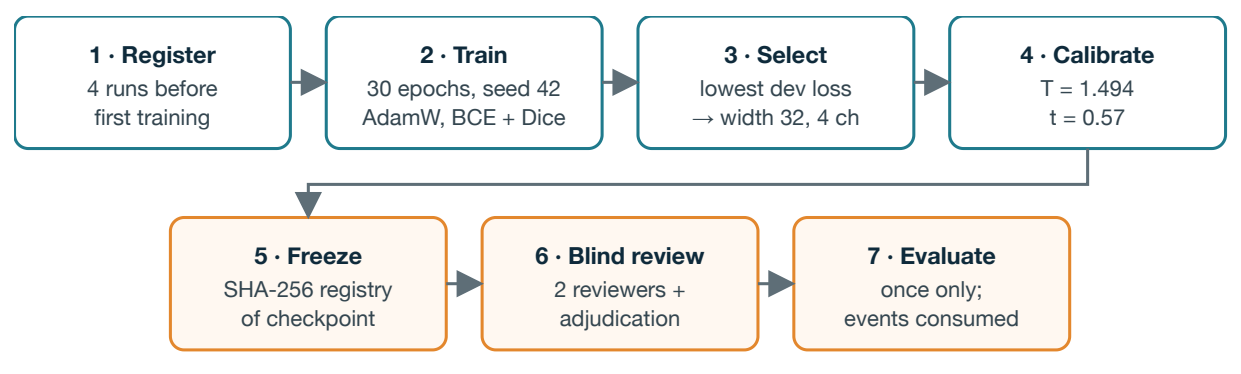
Kuro Siwo labels only, put again on CDSE gamma0 at 20 m. After the registered checks, these chips remained:

TRAIN · 310 chips
Houston 109 · Togo 126 · Peru 75

DEV · 134 chips
Larkana 125 · Illinois 9

BLIND TEST · 3 Himalayan floods
Teesta 2023 · Melamchi 2021 · Chamoli 2021

Registered pipeline (protocol commit bda5748)



1. **1 · Register** — 4 runs before first training
2. **2 · Train** — 30 epochs, seed 42, AdamW, BCE + Dice
3. **3 · Select** — lowest dev loss → width 32, 4 ch
4. **4 · Calibrate** — T = 1.494, t = 0.57
5. **5 · Freeze** — SHA-256 registry of checkpoint
6. **6 · Blind review** — 2 reviewers + adjudication
7. **7 · Evaluate** — once only; events consumed

Loss: BCE (positive weight 2) + Dice. **Optimiser:** AdamW, learning rate 3e-4, batch 4, 30 epochs, seed 42, mixed precision. Four registered variants: width 16 or 32 × 4 or 6 channels.

Training data: Kuro Siwo (Bountos et al., NeurIPS Datasets and Benchmarks 2024).

10 SELECTION AND CALIBRATION

All four registered runs pass the precision floor

$p = \sigma(z / T),\ T = 1.494$ (fitted on dev)
$t^* = \operatorname{argmin}_t 5 \cdot \mathrm{FN}(t) + \mathrm{FP}(t)$
subject to $\mathrm{Precision}(t) \ge 0.80$
$t^* = 0.57$ (99 candidate thresholds)

5 : 1 — cost of a missed flood pixel against a false alarm

≥ 0.80 — precision floor, set on development data

A missed flooded road costs a rescuer five times more than a false alarm.

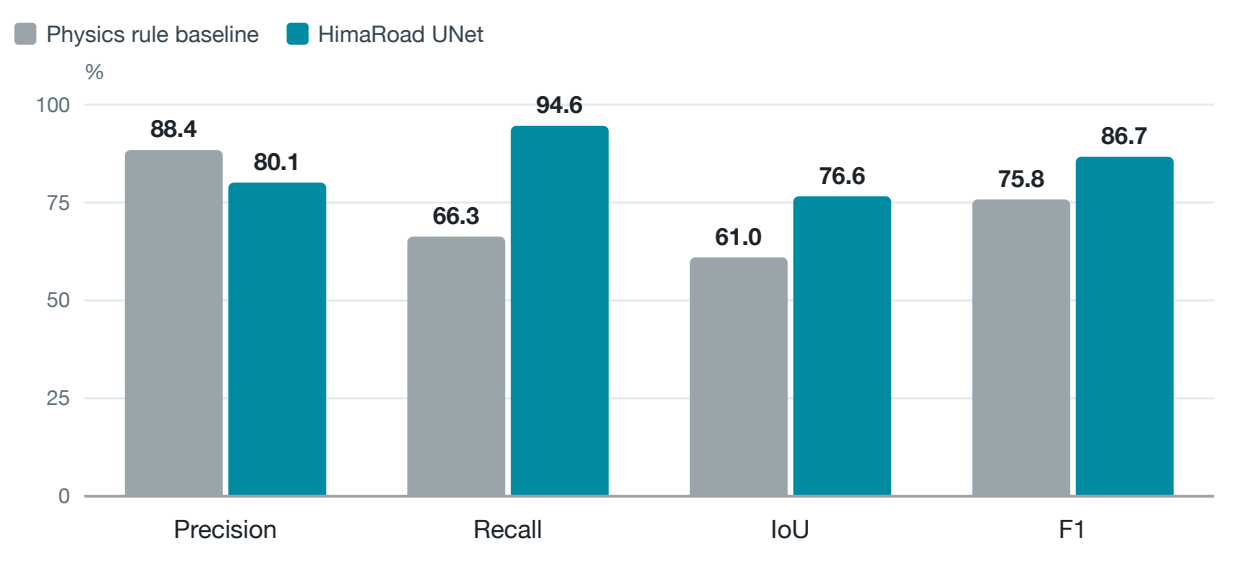
REGISTERED RUN	CALIBRATED DEV IOU
width 32 / 6 channels	75.54%
width 32 / 4 channels (selected)	**76.56%**
width 16 / 6 channels	76.18%
width 16 / 4 channels	75.34%

Selection by lowest best dev loss (0.4228). Temperature scaling after Guo et al., ICML 2017.

Reproducible. A CPU replay matched every calibration table. Max dev-loss difference: 4.5e-6. Temperature 1.4940904 and threshold 0.57 were fixed before the Himalayan test.

11 DEVELOPMENT RESULTS

Development F1 86.7%, against 75.8% for the rule baseline



94.6% — recall: flooded pixels found (baseline 66.3%)

+15.6 pts — IoU gain: 0.7656 against 0.6096

METHOD	PREC.	RECALL	IOU	F1
Rule baseline	0.8835	0.6629	0.6096	0.7575
Selected UNet	**0.8008**	**0.9457**	**0.7656**	**0.8672**

1,167,702 valid development pixels (Larkana + Illinois). Checkpoint 3ed60130.

Read with care. The same two lowland events also selected the model and its threshold. These numbers are development evidence, not independent accuracy, and Larkana dominates the pixel counts.

12 GENERALISATION TEST

A blind, once-only test on held-out Himalayan floods

1. **Pre-select** Teesta 2023, Melamchi 2021 and Chamoli 2021 before training. Never tuned on.
2. **Freeze** predictions and inputs, pinned by SHA-256. A re-freeze was bit-identical.
3. **Blind labels:** two reviewers label in a label station. No prediction is visible.
4. **Adjudicate** together. Only disagreement pixels change.
5. **Evaluate once.** The registry consumes each event.

EVENT	PX	UNET R / IOU	BASELINE P / R / IOU	K
Teesta 2023	81	0 / 0	0.688 / 0.136 / 0.128	0.000
Melamchi 2021	970	0 / 0	1.000 / 0.013 / 0.013	0.679
Chamoli 2021	247	0 / 0	undef. / 0 / 0	0.102

0 / 1,298 — labelled new-water pixels found by the UNet (recall 0, IoU 0; precision undefined)

24 / 1,298 — found by the rule baseline: precision 0.828, recall 0.018

The model does not yet transfer to narrow Himalayan valleys. No threshold, model or checkpoint changed after the test. Next version: mountain-domain training data.

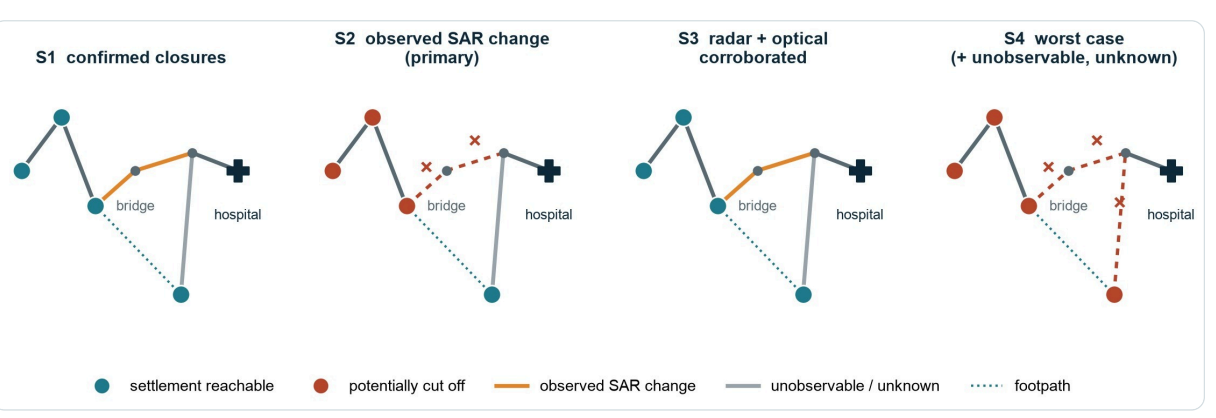
One valley-enriched 128×128 window per event: small, not representative. Reviewer agreement was low on Teesta and Chamoli. In six secondary windows neither method found labelled new water. Result 63a3e785 · freeze a0a70c07 · review bdfca70e.

D What was damaged, who is cut off, and how good is it?

From hazard pixels to settlement access, then checked against a reference product and live drills

13 FROM HAZARD TO ACCESS

Four hypothetical scenarios separate evidence from assumption



SCENARIO	LINKS REMOVED FROM THE GRAPH	ROAD-CUT
S1	Independently sourced confirmed closures	0
S2 (primary)	**S1 + links with observed change or likely impact**	**22 of 73**
S3	S1 + links with radar and optical corroboration (subset of S2)	0
S4 (worst)	S2 + possible-impact, unobservable and unknown links	45 of 73

Directed graphs
Vehicle and pedestrian graphs with access and one-way tags.

Settlements
Clusters of ≥ 10 OSM buildings within 50 m; 47 of the 73 here are clusters.

Baseline first
Access is checked before the event, so a place that never had a road is not a new cut-off.

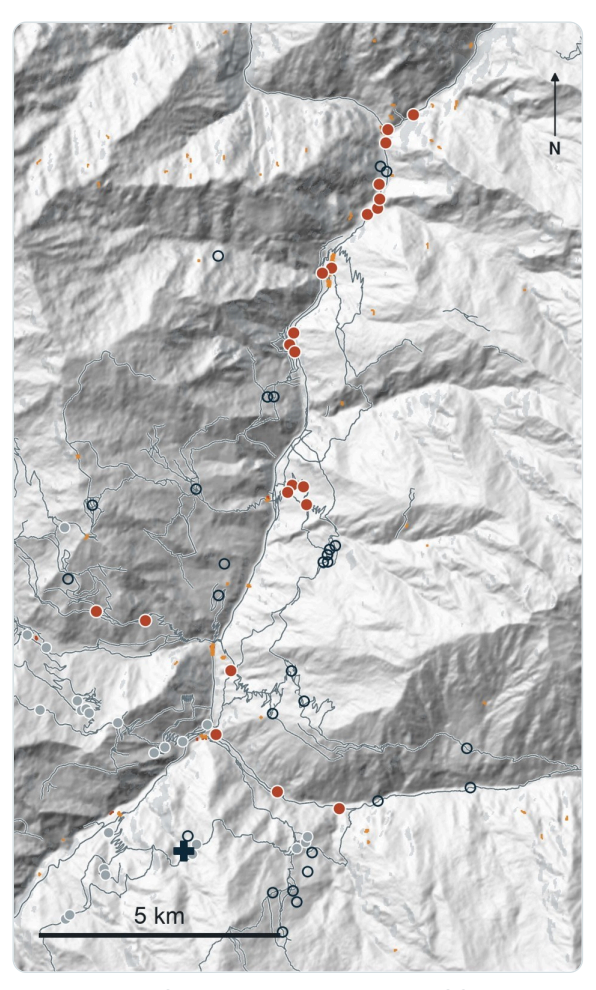
Reverse Dijkstra
Multi-source search from every destination: nearest town and health facilities.

Exposure states. Each road link gets one state: potentially exposed, observed with no detected change, or unobservable. Operational status stays `unknown` unless an independent source gives it.

A candidate route is **mapped reachability, not a safe route**.

14 REAL RESULT · TRISHULI · SCENARIO 2

22 of 73 settlements can lose road access



22 / 73 — potentially road-cut in S2 (observed SAR change)

23 — access uncertain: cut only if unknown links fail (S4)

14 of 22 — road-cut settlements still have a mapped trail

121 — suspected affected links, all from SAR evidence

102 — potentially exposed buildings

28 — had no mapped road access before the event

0 — primary-care disconnections in S2

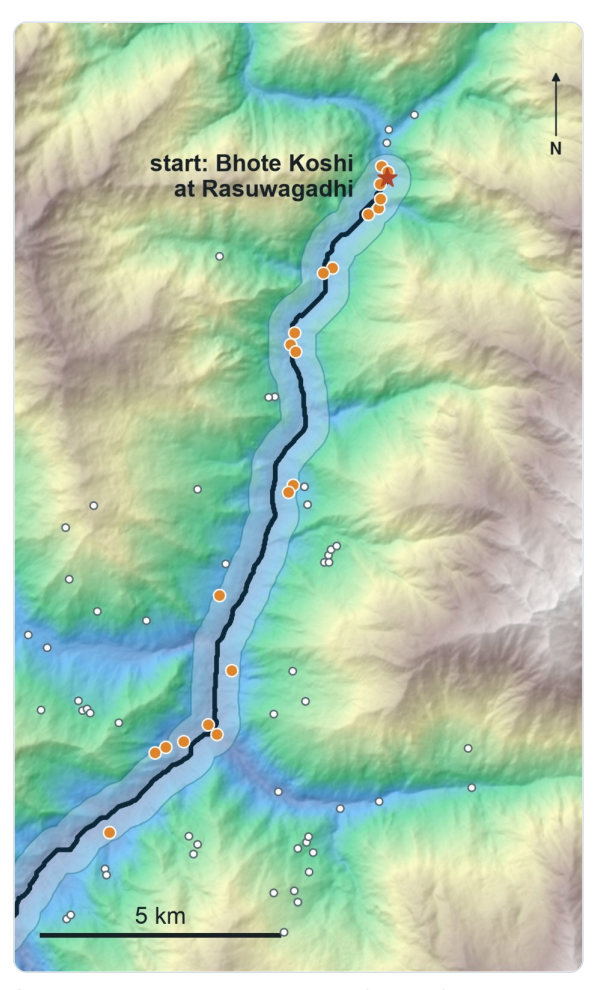
Hillshade from Copernicus DEM, pre-event OSM roads, 5 km scale bar.

● potentially road-cut (22) ● access uncertain (23) ○ no road before the event (28)
+ destination (town or care) ▬ suspected affected link

Hypothetical screening. Overlap means potentially affected.

15 BONUS · DEM FLOW PATH

Tracing where the water goes next



27.0 km — D8 steepest-descent path from the Bhote Koshi at Rasuwagadhi

21 — settlements inside the 500 m corridor, in downstream order

1,122 — DEM cells on the path

● in the corridor ○ other settlement

Start: Bhote Koshi at Rasuwagadhi (red star). Black line: D8 path. Blue band: 500 m corridor.

Method. Priority-flood conditioning of the Copernicus GLO-30 DEM (Barnes 2014), then D8 steepest descent from the start point.

In the web app: click the map, then "Trace flow path". The app lists the settlements along the path.

Every output carries the label **"DEM-derived flow path, not an inundation forecast"**. It shows where water from the point goes on the terrain. It gives no depth, timing or extent.

16 CASE STUDY VS REFERENCE

Frozen case study compared with Copernicus EMSR927

EMSR927 AOI	AREA RECALL	TRANSPORT RECALLED	"NO DAMAGE" FLAGGED
AOI01 Syapru Besi	**0.012**	**6 of 19**	0 of 8
AOI02 Timure	**0.038**	**7 of 13**	none graded
AOI03 Bidur, AOI05 Phosretar			unavailable: outside our AOI
AOI04 Bharatpur, AOI06 Kyundi			unavailable: reference not feasible

- **Two runs frozen first** (rule baseline and UNet, producer 23a2527), before EMSR927 was opened. They are identical on every scored item: the UNet's one water candidate was removed by the minimum-mapping-unit and permanent-water step. The flood extent is empty in both, so the disturbance layer is **rule evidence, not model output**.
- **What we compared.** EMSR927's only observed-event class here is "Landslide", so water IoU is unavailable, not zero. We use positive-only area recall of the footprint, and graded transport features on potentially exposed links (20 m match).
- **Failure analysis.** We observed most of the footprint (93.1% in AOI01, 76.1% in AOI02), so the low recall is a **detection miss, not a visibility gap**. AOI01: 11 of 13 misses lie on links observed with no detected change. AOI02: 5 of 6 misses are fully or partly unobservable.
- The reference reads very-high-resolution optical images by eye. We measure 20 m radar change.

Result: the map shows where links are potentially exposed. It **does not reliably delineate this landslide event**. Overlap is agreement between two satellite products; it never confirms damage or closure.

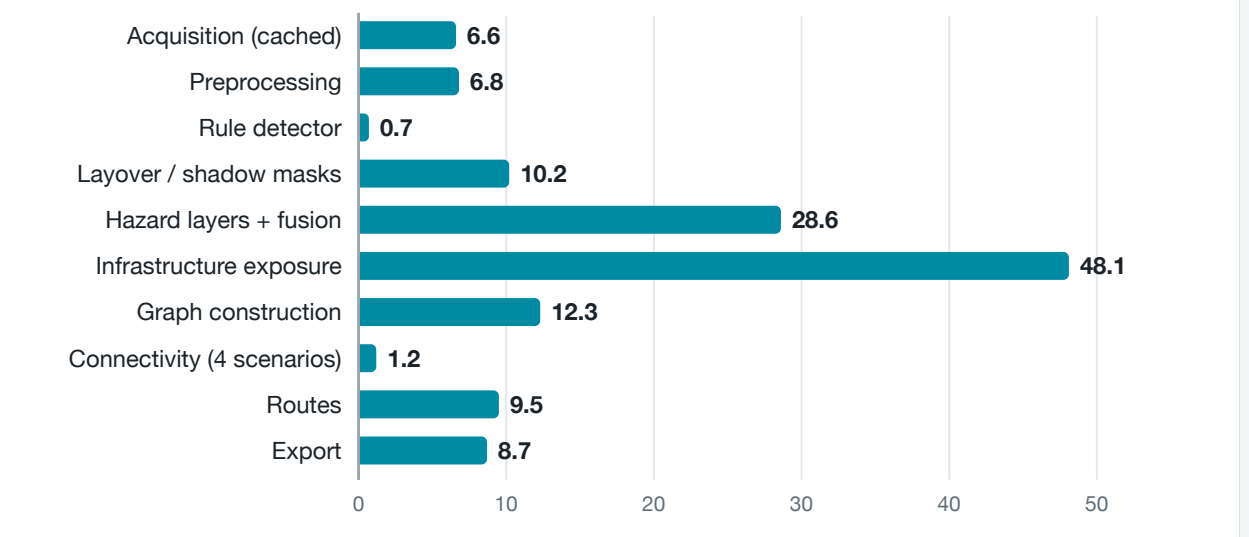
17 LIVE ROBUSTNESS AND SPEED

Six worldwide drills: two complete, four honest stops

REGISTERED AOI	OUTCOME
Lokoja, Nigeria	complete cold run 99.7 s; later 56.7 s, 59.9 s
Lismore, Australia	complete runs 45.9–69.7 s
Valencia, Spain	timed out at the 180 s budget
Porto Alegre, Brazil	incomplete no same-track pre-event scene
Manila 2010	incomplete before Sentinel-1
Germany, 7,807 km²	aoi_too_large nothing downloaded

UNet detector, 180 s budget, AOIs outside training, development and test. Robustness only, no accuracy claim. One Lokoja rerun took 159.8 s: the Sentinel-2 search is not cached.

Stage time, frozen Trishuli run (s), total 133 s



1.7 s — UNet on 600 km², GPU (4.9 s CPU); detector only

28.6× — faster connectivity on Melamchi: 423.5 → 14.8 s

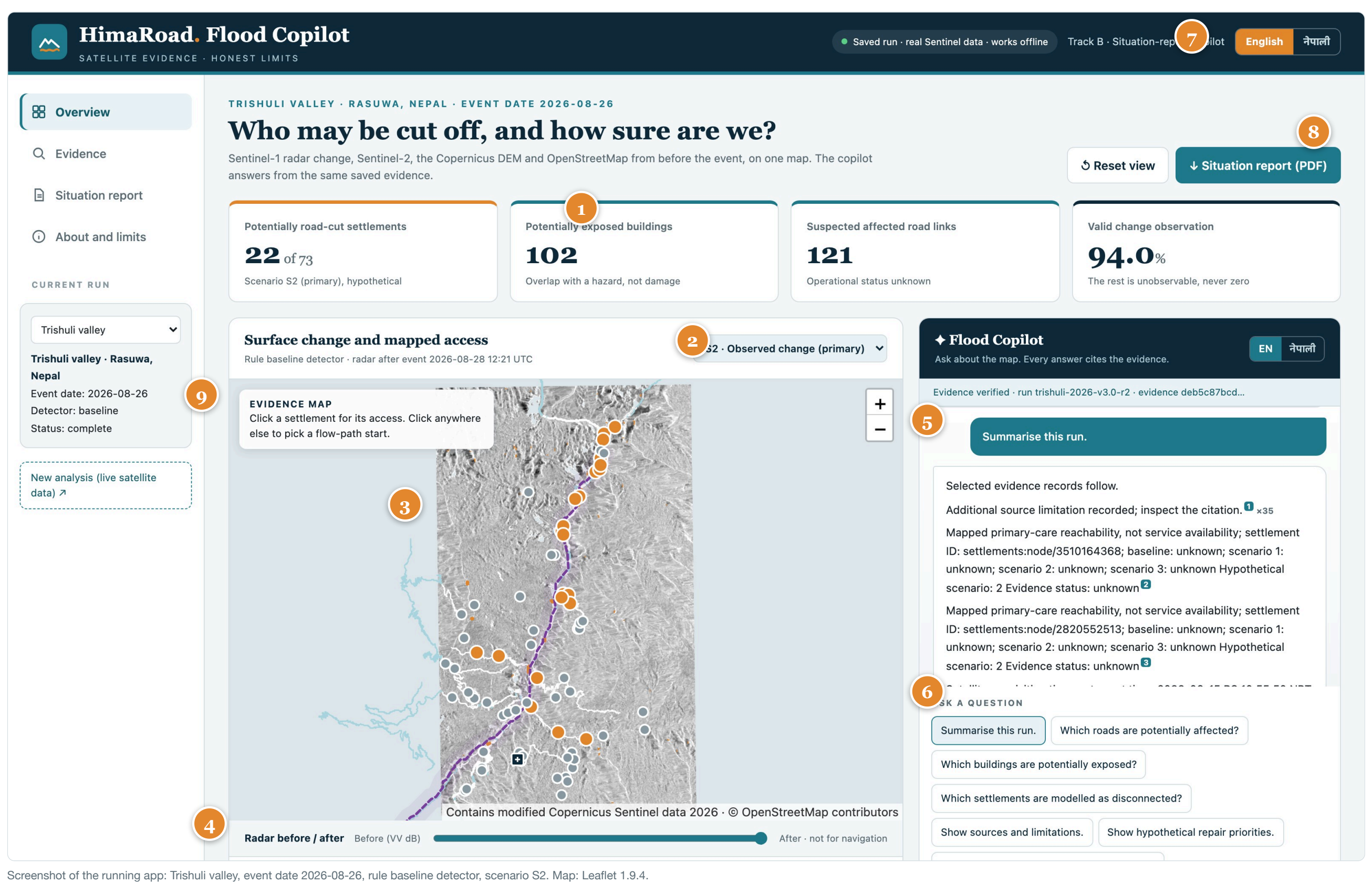
518 s — Valencia (dense city) diagnostic

E Our AI, part 2: the situation-report copilot and the web app

What a rescuer opens, asks and prints

18 THE PRODUCT

One web app: map dashboard and copilot on one page, in English and Nepali


HimaRoad. Flood Copilot
SATELLITE EVIDENCE · HONEST LIMITS
Saved run · real Sentinel data · works offline
Track B · Situation-report copilot
English
नेपाली
Overview
Evidence
Situation report
About and limits
CURRENT RUN
New analysis (live satellite data) ↗
Reset view
↓ Situation report (PDF)
S2 · Observed change (primary)
+
−
EN
नेपाली
ASK A QUESTION
1
2
3
4
5
6
7
8
9

TRISHULI VALLEY · RASUWA, NEPAL · EVENT DATE 2026-08-26

Who may be cut off, and how sure are we?

Sentinel-1 radar change, Sentinel-2, the Copernicus DEM and OpenStreetMap from before the event, on one map. The copilot answers from the same saved evidence.

Trishuli valley

Trishuli valley · Rasuwa, Nepal
Event date: 2026-08-26
Detector: baseline
Status: complete

Potentially road-cut settlements	Potentially exposed buildings	Suspected affected road links	Valid change observation
22 of 73	102	121	94.0%
Scenario S2 (primary), hypothetical	Overlap with a hazard, not damage	Operational status unknown	The rest is unobservable, never zero

Surface change and mapped access
Rule baseline detector · radar after event 2026-08-28 12:21 UTC

EVIDENCE MAP
Click a settlement for its access. Click anywhere else to pick a flow-path start.

Contains modified Copernicus Sentinel data 2026 · © OpenStreetMap contributors

Radar before / after — Before (VV dB) — After · not for navigation

✦ Flood Copilot
Ask about the map. Every answer cites the evidence.

Evidence verified · run trishuli-2026-v3.0-r2 · evidence deb5c87bcd...

Summarise this run.

Selected evidence records follow.

Additional source limitation recorded; inspect the citation.[1] ×35

Mapped primary-care reachability, not service availability; settlement ID: settlements:node/3510164368; baseline: unknown; scenario 1: unknown; scenario 2: unknown; scenario 3: unknown Hypothetical scenario: 2 Evidence status: unknown[2]

Mapped primary-care reachability, not service availability; settlement ID: settlements:node/2820552513; baseline: unknown; scenario 1: unknown; scenario 2: unknown; scenario 3: unknown Hypothetical scenario: 2 Evidence status: unknown[3]

Summarise this run. · Which roads are potentially affected? · Which buildings are potentially exposed? · Which settlements are modelled as disconnected? · Show sources and limitations. · Show hypothetical repair priorities.

Screenshot of the running app: Trishuli valley, event date 2026-08-26, rule baseline detector, scenario S2. Map: Leaflet 1.9.4.

1. **Metric cards** for the saved run. The road-cut count changes with the scenario.
2. **Scenario menu** (S1 to S4). The settlement table and the map markers follow it.
3. **Evidence map:** the run's own Sentinel-1 image, hazard layers, roads coloured by evidence class, exposed buildings, settlements and hospitals. Layer boxes show and hide each one.
4. **Radar before / after slider** compares the images from before and after the event.
5. **Flood Copilot:** each answer has numbered citations and a count of cited evidence claims.
6. **Predefined questions.** A typed question goes to the question that it matches.
7. **English / नेपाली** switch changes the whole page and the copilot language.
8. **Situation report (PDF)** in English or Nepali, ready to download.
9. **Current run** selector: every saved bundle (Trishuli 2026, Melamchi 2021).

Four pages

Overview
Map, settlement table, copilot, flow-path tool. Click a settlement for its access, exposed buildings and nearest mapped hospital.

Evidence
Scene IDs, observation gaps, OSM snapshot, thresholds, detector and EMSR927 comparison, scenarios, provenance.

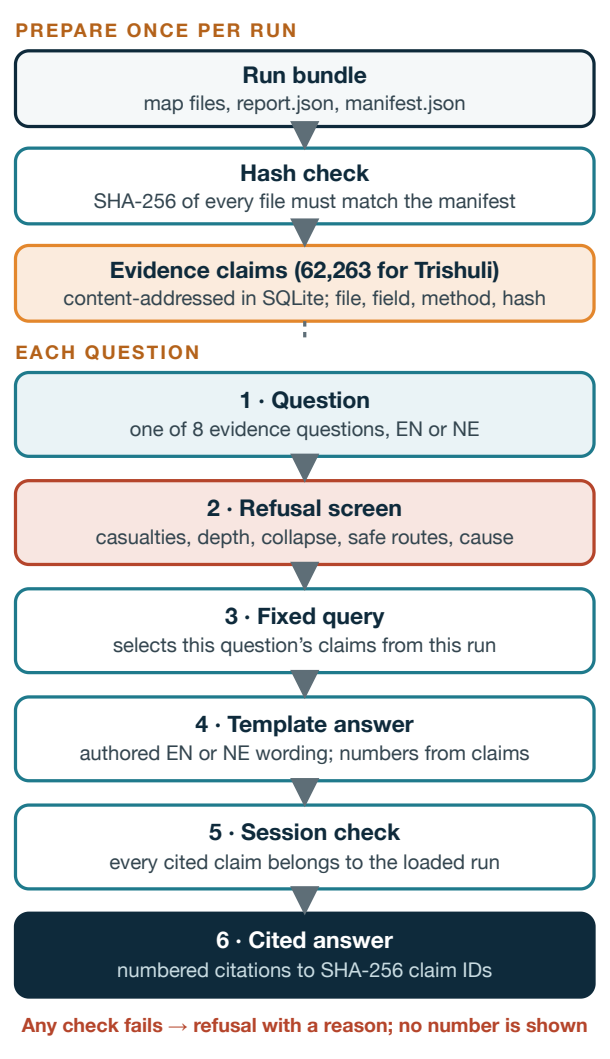
Situation report
"Prepare English PDF" or the Nepali button makes the one-page report.

About and limits
Measured results, the limits of the system and the attributions.

Saved runs work without a network connection. The copilot loads the run and checks its file hashes in about 15 s.

19 SITUATION-REPORT COPILOT

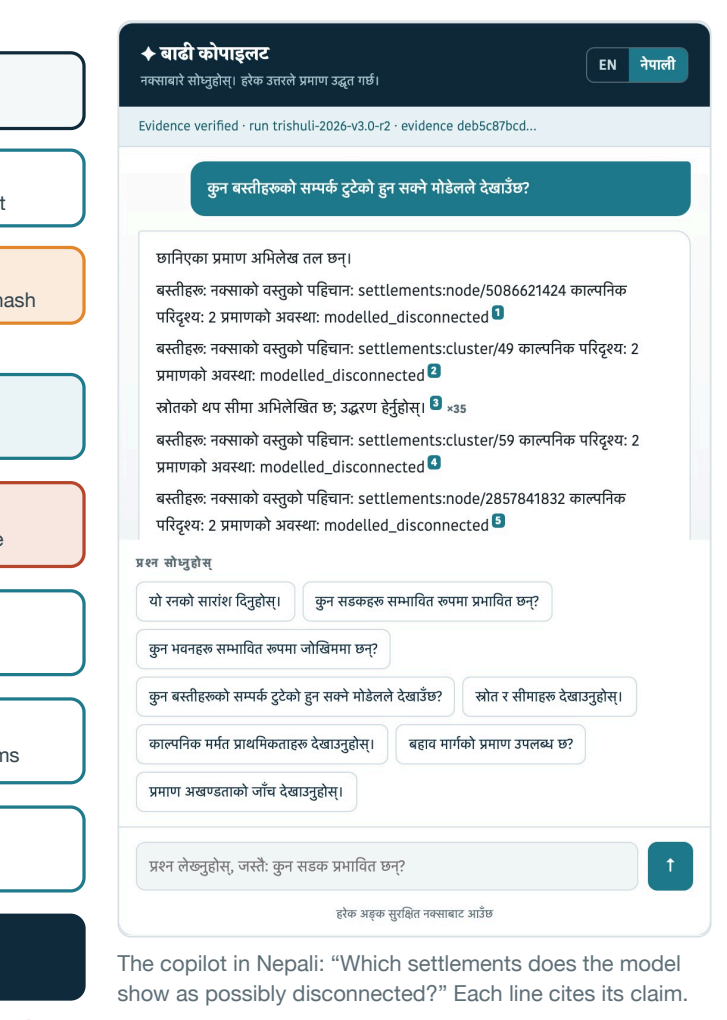
A copilot that cannot invent a number



PREPARE ONCE PER RUN
- **Run bundle** — map files, report.json, manifest.json
- **Hash check** — SHA-256 of every file must match the manifest
- **Evidence claims (62,263 for Trishuli)** — content-addressed in SQLite; file, field, method, hash

EACH QUESTION
1. **1 · Question** — one of 8 evidence questions, EN or NE
2. **2 · Refusal screen** — casualties, depth, collapse, safe routes, cause
3. **3 · Fixed query** — selects this question's claims from this run
4. **4 · Template answer** — authored EN or NE wording; numbers from claims
5. **5 · Session check** — every cited claim belongs to the loaded run
6. **6 · Cited answer** — numbered citations to SHA-256 claim IDs

Any check fails → refusal with a reason; no number is shown



✦ बाढी कोपाइलट
नक्साबारे सोध्नुहोस्। हरेक उत्तरले प्रमाण उद्धृत गर्छ।

Evidence verified · run trishuli-2026-v3.0-r2 · evidence deb5c87bcd...

कुन बस्तीहरूको सम्पर्क टुटेको हुन सक्ने मोडेलले देखाउँछ?

छानिएका प्रमाण अभिलेख तल छन्।
बस्तीहरू: नक्साको वस्तुको पहिचान: settlements:node/5086621424 काल्पनिक परिदृश्य: 2 प्रमाणको अवस्था: modelled_disconnected[1]
बस्तीहरू: नक्साको वस्तुको पहिचान: settlements:cluster/49 काल्पनिक परिदृश्य: 2 प्रमाणको अवस्था: modelled_disconnected[2]
स्रोतको थप सीमा अभिलेखित छ; उद्धरण हेर्नुहोस्।[3] ×35
बस्तीहरू: नक्साको वस्तुको पहिचान: settlements:cluster/59 काल्पनिक परिदृश्य: 2 प्रमाणको अवस्था: modelled_disconnected[4]
बस्तीहरू: नक्साको वस्तुको पहिचान: settlements:node/2857841832 काल्पनिक परिदृश्य: 2 प्रमाणको अवस्था: modelled_disconnected[5]

प्रश्न सोध्नुहोस्
यो रनको सारांश दिनुहोस्। · कुन सडकहरू सम्भावित रूपमा प्रभावित छन्? · कुन भवनहरू सम्भावित रूपमा जोखिममा छन्? · कुन बस्तीहरूको सम्पर्क टुटेको हुन सक्ने मोडेलले देखाउँछ? · स्रोत र सीमाहरू देखाउनुहोस्। · काल्पनिक मर्मत प्राथमिकताहरू देखाउनुहोस्। · बहाव मार्गको प्रमाण उपलब्ध छ? · प्रमाण अखण्डताको जाँच देखाउनुहोस्।

प्रश्न लेख्नुहोस्, जस्तै: कुन सडक प्रभावित छन्?

हरेक अङ्क सुरक्षित नक्साबाट आउँछ

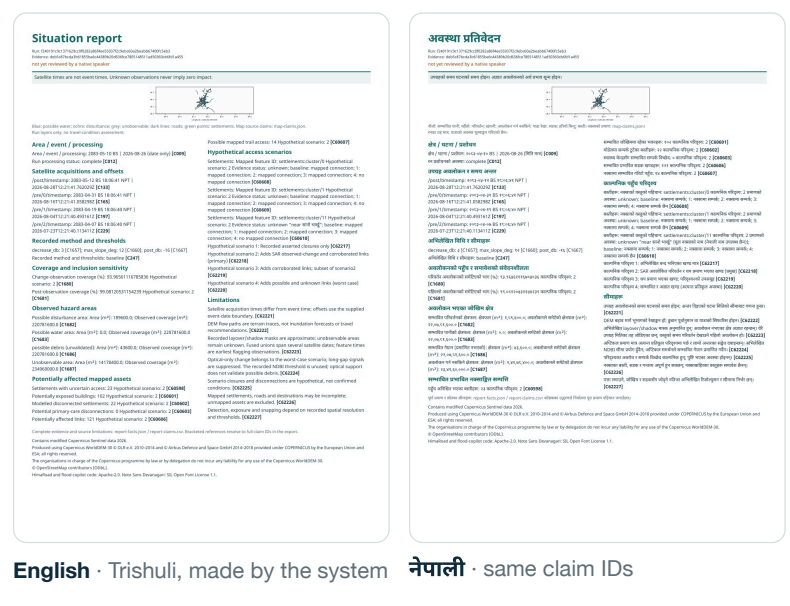
The copilot in Nepali: "Which settlements does the model show as possibly disconnected?" Each line cites its claim.

- **8 evidence questions:** run summary, affected roads, exposed buildings, disconnected settlements, sources and limitations, hypothetical repair priorities, flow-path evidence, evidence integrity.
- **It refuses, with a reason:** casualty or missing-person questions, water depth and forecasts, confirmed collapse, destruction or closure, safe routes and the cause of the event, questions outside the loaded run. **Missing observations stay unknown.**
- **Deterministic:** predefined questions and authored templates. No number is ever generated; each one is copied from a claim.

SAVED RUN	TRISHULI	MELAMCHI
Claims verified	62,263	178,648
Offline preparation	67.57 s	524.46 s
Predefined answer	0.20–0.35 s	0.62–1.02 s
Final PDF, EN / NE	8.15 / 7.53 s	45.01 / 53.58 s

20 BILINGUAL REPORTS

One-page situation reports a field team can print



English · Trishuli, made by the system **नेपाली** · same claim IDs

32 / 32 — cases (8 types × EN/NE × 2 areas) with exact claim-ID parity

165 — Nepali wording entries reviewed by a native speaker: 162 approved, 3 corrected

4 / 4 — proof PDFs on one page (Trishuli, Melamchi; EN, NE)

- **Rendering:** WeasyPrint + HarfBuzz shaping + Noto Sans Devanagari, shaping checked in tests. Correct glyphs, native-review marker, all 8 authored caveats, full credits.
- **Nepali conventions:** Devanagari numerals, lakh grouping, Bikram Sambat dates. Run records (scene IDs, thresholds) stay in English.
- **Exports:** JSON and CSV of every claim with its source hash; offline map.html and KML for field devices.

One real claim from the Trishuli run

EN · Potentially exposed buildings: 102 Hypothetical scenario: 2 **[C60601]**
NE · सम्भावित जोखिममा रहेका भवनहरू: १०२ काल्पनिक परिदृश्य: 2 **[C60601]**

id	`60ef6d96…18740a2445` (SHA-256)
value	102 (count) · scenario 2, hypothetical · status unknown
source	`report.json /summary/potentially_exposed_buildings`
sha256	`15350b3c…a1903b75` (hash of report.json)
method	upstream-field · schema 3.0 · producer commit 23a2527

The bracketed reference resolves to the full claim ID in the export. For a UNet run, model identity also becomes claims: model version, checkpoint SHA-256, temperature and threshold.

F Trust, limits, next steps, and how to run it

Provenance, responsible wording, what the system cannot do, the roadmap and the commands

21 REPRODUCIBILITY AND RESPONSIBLE DESIGN

Every number traces back to a file, a commit and a hash

SHA-256 — for every file; a changed file stops the import

3.0 — versioned output contract between the two systems

23a2527 — producer commit of the frozen Trishuli runs (Melamchi 7c1b74a)

Every run records

producer_commit · schema_version 3.0 · scene IDs + timestamps · thresholds + hazard_version · model_version + checkpoint SHA-256 · OSM snapshot + edit proof · input SHA-256 checksums · timings per stage

Words that a rescuer can trust

THE SYSTEM SAYS	THE SYSTEM NEVER SAYS
"Potentially exposed" or "potentially affected"	Destroyed, collapsed or closed
"Unobservable" = unknown, never zero	Casualty figures
Status "unknown" unless independently sourced	The cause of the event
Scenario results labelled hypothetical	"Safe route"
Satellite time kept apart from event time	(and shows no images of victims)

HimaRoad is an **educational research prototype**. It supports the people who decide. It does not replace local knowledge or official assessments.

22 LIMITATIONS

What the system cannot do

- **No early warning.** Sentinel satellites revisit every few days. The system maps what the next pass shows, after the event; it cannot warn of a glacier collapse minutes ahead.
- **Unobservable areas.** 24.9% of the AOI is masked and 56.1% of road links are unobservable. Unknown, not safe, not zero.
- **Exposure is not damage.** No confirmed destruction, collapse or closure. Operational status stays unknown.
- **Scenarios are hypothetical.** Routes are mapped reachability. OSM gaps, tags and the finite graph buffer limit results.
- **Himalayan transfer.** The UNet found 0 of 1,298 labelled new-water pixels on the blind test.
- **No validated debris class.** The UNet maps water only; `possible_debris` is unvalidated rule evidence (landslide area recall 0.012 and 0.038).
- **Sensing limits.** Mixed pixels, vegetation, snow, optical cloud gaps, features narrower than about 20 m.
- **Dense cities are slow.** Valencia: 518 s against 180 s (exposure 246.9 s, graph 126.6 s, export 72.6 s).
- **Data gaps.** No same-track pair or a date before Sentinel-1 → `incomplete`; above 600 km² → `aoi_too_large`.
- **Copilot.** No route reopening above 50 settlements or 2,000 links. The map view shows at most 100 features per layer; full exports stay authoritative.

No casualty figures, no cause, no confirmed collapse, no safe route. Independent inspection is necessary before any use.

23 IMPACT AND ROADMAP

From prototype to a tool in the hands of relief teams

1. **Integrate** — One hosted dashboard: map, chat and bilingual report on one screen.
2. **Mountain model** — Terrain-aware inputs and more mountain examples. A new blind Himalayan test.
3. **Debris class** — Turn "possible debris" into an expert-reviewed class.
4. **Field-ready** — Offline bundle, faster dense-city runs, measured end-to-end latency.
5. **More languages** — Hindi and other languages of the Himalayan region.

Goal: when a river changes in less than an hour, a relief team knows a little sooner where to go.

Deliverables

- **Map dashboard:** the web app in `web/`, port 8000
- **Situation reports:** EN + NE PDF; Trishuli examples in `results/situation-report/`
- **Report:** REPORT.md and a designed 6-page REPORT.pdf, with limitations
- **Deck:** 15 slides (PDF); the PowerPoint file is in the v1.0 release
- **Case study vs EMSR927:** REPORT §5 and `results/emsr927/`
- **How to run:** README

24 RUN IT

Install, download the saved runs, start with one command

```
# 1 · get the code and the locked environment (Python 3.12, uv), with the UNet
git clone https://github.com/G-Shack/HimaRoad_flood-copilot.git && cd HimaRoad_flood-copilot
uv sync --locked --extra unet
# 2 · pinned Nepali font, then the saved runs and UNet weights (every SHA-256 checked)
.venv/bin/python scripts/copilot/fetch_font.py --output data/copilot/cache/fonts
REPO=G-Shack/HimaRoad_flood-copilot scripts/get_data.sh
# 3 · prepare the case study for the copilot (about 40 s and 12 GB RAM)
python -m scripts.copilot.prepare "$HIMAROAD_DATA_DIR/bundles/trishuli-2026-v3.0-r2" --output data/copilot/trishuli.json
# 4 · start the web app (:8000) and the analysis workbench (:8501)
scripts/start.sh            # then open http://127.0.0.1:8000/?run=trishuli-2026-v3.0-r2
# new live analysis: free CDSE (Sentinel Hub) and ohsome accounts in .env
himaroad precheck --bbox 85.30,28.12,85.42,28.30 --event-date 2026-08-26
himaroad run --provider cdse --bbox 85.30,28.12,85.42,28.30 --event-date 2026-08-26 --output .../my-run \
             --detector unet --model-config configs/model_unet.yaml
himaroad bundle --run .../my-run --out "$HIMAROAD_DATA_DIR/bundles/my-run"
```

PART	FOLDER	WHAT IT GIVES
HimaRoad	`src/himaroad/` `configs/`	UNet, hazard layers, cut-off analysis, flow path, map HTML, PDF, GeoJSON, KML
flood-copilot	`backend/` `upstream/`	EN/NE questions and answers, one-page report; every number from saved maps
Web app	`web/`	Map, copilot, evidence and report on one port (8000)
Workbench	`app.py` `views/` `ui/`	Streamlit pages that start a new live analysis (8501)

Needs: macOS or Linux, Python 3.12 and uv, about 24 GB RAM to prepare the case study, and Pango for the PDFs.

Saved runs: Trishuli 2026 (rule baseline and UNet) and Melamchi 2021. They work offline.

Sources: one repository with clean copies of HimaRoad `2c2619f` and flood-copilot `8159771`. Each run records its own producer commit.

Attributions and citations

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© OpenStreetMap contributors (ODbL), historical snapshot via ohsome. European Union, Copernicus Emergency Management Service data (EMSR927), validation only, after predictions were frozen.

References

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HimaRoad Flood Copilot

Team HimaRoad · github.com/G-Shack/HimaRoad_flood-copilot

Multimodal AI Hackathon 2026 · Track B: Mapping Flood Damage from Space

Code: Apache-2.0. Data, model weights and reference products keep their own terms. Leaflet 1.9.4 (BSD-2-Clause). Noto Sans Devanagari (SIL Open Font License).

Educational research prototype, not an operational tool.